

Ultrawideband Wavelength Conversion Using Cascaded SOA-Based Wavelength Converters

Motoharu Matsuura, Naoto Kishi, *Member, IEEE*, and Tetsuya Miki, *Fellow, IEEE*

Abstract—An all-optical wavelength converter with a large wavelength hopping range is proposed and demonstrated. This converter consists of multistage cascaded wavelength converters using semiconductor optical amplifiers each with different gain band. Each of the cascaded wavelength converters enables us to perform both noninverted (NIV) and inverted (IV) operations. Conversion performance is compared at NIV and IV operations in terms of static characteristics as a function of input/output power of the converter. While good conversion performances are achieved at both operations, the IV wavelength conversion has better cascading ability to obtain a high-quality converted signal for the cascaded scheme. Moreover, signal amplitude regeneration is demonstrated by repeating the IV wavelength conversion. Finally, we successfully demonstrate, for the first time, ultrawideband wavelength conversion, including over 300-nm wavelength hopping to the shorter wavelength side with a triple-stage cascaded wavelength converter.

Index Terms—All-optical devices, broadband wavelength conversion, optical fiber communication, optical regeneration, semiconductor optical amplifiers (SOAs), wavelength division multiplexing (WDM).

I. INTRODUCTION

ALL-OPTICAL wavelength converters (AOWCs) must play a significant role in future wavelength-division-multiplexing (WDM) transmission system, because they avoid wavelength blocking and improve utilization of signal wavelengths [1], [2]. To provide a variety of communication services to larger network areas and individual clients, there has been a growing interest in ultrawideband photonic networks utilizing the entire low-loss bandwidth of silica optical fiber [3]. In such networks, various WDM technologies such as over 1000-channel WDM transmission [4] and metro/access wide passband WDM [5] will be required, and a vast number of available wavelengths must be controlled and managed in the ultrawideband wavelength region. Therefore, it is very important to develop and implement broadband AOWCs, which hop flexibly an arbitrary wavelength in such a wide wavelength range. Many approaches have been demonstrated utilizing nonlinearities in optical fibers or semiconductor optical amplifiers (SOAs) so far.

Although fiber-based AOWCs are attractive for ultrafast signal processing owing to the instantaneous response of Kerr nonlinearity in fibers, these wavelength hopping ranges are strongly restricted by phase mismatch and walk off induced by the group-velocity difference between wavelengths of in-

teraction signals. To ease such obstacles, many AOWCs using various fibers have been reported [6]–[11]. Recently, over 193-nm wavelength conversion by a special fiber improving higher order dispersion performance has been demonstrated [12]. However, further expansion of the hopping range requires higher input optical power, as well as more complicated control of fiber parameters.

SOA-based AOWCs have some advantages such as low-power consumption and compact size [13]. A number of approaches based on various nonlinear effects have already been demonstrated [14]–[16]. In particular, cross-phase modulation (XPM) is very useful for broadband wavelength conversion, since XPM in SOAs is effective in the wide wavelength range. Lacey *et al.* have demonstrated a large wavelength hopping from 1300 to 1550 nm using a single 1.3 μm SOA [17]. This indicates that XPM is useful for broadband down-conversion (conversion of short-to-long wavelength). In contrast, it is very difficult to realize broadband up-conversion (conversion of long-to-short wavelength) with such large wavelength hopping. Because SOAs give rise to a high absorption effect based on its energy band gap at a much shorter wavelength outside of the gain band, the converted signal suffers from high absorption loss, and its power level is drastically degraded. Therefore, these wavelength hopping ranges are mainly restricted by the gain bandwidth of the SOA.

To overcome the above issue, we previously proposed a novel broadband-wavelength-conversion scheme using a multistage cascaded SOA-based wavelength converter [18]. In this scheme, we employed a dual-stage cascaded wavelength converter based on delayed-interferometric switches each with a different gain band of the SOAs, and demonstrated up to 160-nm up-conversion for the first time. Moreover, we successfully achieved error-free operations for various wavelength hopping ranges of 60–230 nm [19]. However, in order to avoid a patterning effect and obtain a sufficiently high switching energy for wavelength conversion, special 2.5-Gb/s return-to-zero signals with low duty ratio less than 0.1 had to be employed. In addition, accumulated signal degradation of the converted signal was observed as the number of cascaded converters was increased.

This paper presents a novel cascaded scheme for improving conversion performance and wavelength hopping range of our proposed method [20]. The scheme consists of multistage AOWCs by using a single SOA, each with a different gain band. These converters enable us to perform both noninverted (NIV) and inverted (IV) wavelength conversions of a nonreturn-to-zero (NRZ) signal. In particular, the IV conversion has good cascading ability to obtain a high-quality signal at a much shorter

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The authors are with the Department of Information and Communication Engineering, University of Electro-Communications, Chofu, Tokyo 182-8585, Japan (e-mail: matsuura@ice.uec.ac.jp; kishi@ice.uec.ac.jp; miki@ice.uec.ac.jp).

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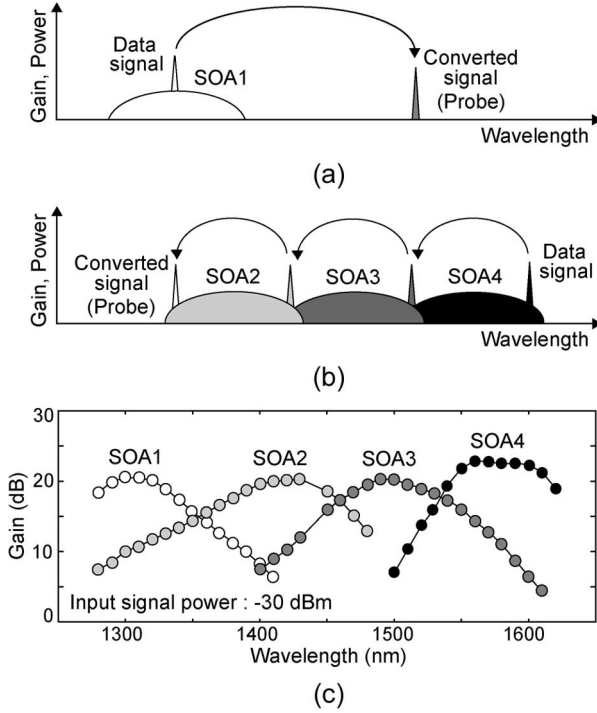


Fig. 1. Cascaded scheme for broadband wavelength conversion. (a) Down-conversion by single-stage wavelength converter using SOA1. (b) Up-conversion by multistage cascaded converter using SOA2, SOA3, and SOA4. (c) Gain characteristics of four SOAs employed in this experiment.

wavelength for the cascaded broadband wavelength conversion. We also realize the amplitude regeneration effect of the input data signal by repeating the IV wavelength conversion. This function helps us to improve the wavelength hopping range of the cascaded scheme. Moreover, we demonstrate ultrawideband wavelength conversion over 300-nm wavelength hopping with low Q -penalties using triple-stage cascaded wavelength converter.

This paper is organized as follows. In Section II, the operation principle of the cascaded scheme for ultrawideband wavelength conversion is presented. In Section III, we describe the configuration of the cascaded SOA-based wavelength converter and investigate its characteristics for the cascaded scheme. The experimental results are presented in Section IV, and its features and advantages are highlighted in the discussion in Section V. The conclusion of this paper is summarized in Section VI.

II. CASCADED SCHEME FOR ULTRAWIDEBAND WAVELENGTH CONVERSION

The cascaded scheme for ultrawideband wavelength conversion with a single or multistage SOA-based wavelength converter is depicted in Fig. 1. In each wavelength converter, the input data signal and continuous-wave (CW) probe beam are launched into an SOA. The input signal encodes the signal information by means of cross-gain modulation (XGM) and XPM onto the CW probe. The wavelengths of the input data and the probe signals are located in sufficiently high gain bandwidth of the SOA if XGM is fully used. On the other hand, XPM can be used to perform wavelength conversion in a wide

wavelength range over the gain band since the SOA's carrier density modulation based on XPM induces a phase modulation, even if the probe wavelength is located at an outside wavelength of the gain bandwidth of the SOA [17]. Thus, in the case of broadband down-conversion, as shown in Fig. 1(a), we obtain a high-quality converted signal using a single SOA-based wavelength converter, which utilizes an SOA with high gain at the input data wavelength.

On the other hand, in the case of broadband up-conversion, the output power of converted signal is drastically degraded due to high absorption loss of the SOA, although the probe light can experience XPM in the SOA. Therefore, as shown in Fig. 1(b), we obtain a high-quality converted signal at a much shorter wavelength by repeating up-conversion within each gain bandwidth [18]. The wavelengths of the input data and probe signals are located at the long and short sides of the gain peak wavelength of the SOA, respectively. In this cascaded scheme, we achieve a flexible large wavelength hopping using a single- or multistage wavelength converter, which employs SOAs with each proper gain band. The proposed scheme is effective in the ultrawide wavelength range without restriction by the hopping range of each converter.

Fig. 1(c) shows the gain characteristics of the employed SOAs. SOA1 is employed for broadband down-conversion, whereas SOA2, SOA3, and SOA4 are employed in each wavelength hopping region for the broadband up-conversion. In these measurements, the input signal powers are set to -30 dBm for each optimized injection current. These SOAs have various gain peak wavelengths and small signal gain over 20 dB in the wide wavelength range.

III. CHARACTERISTICS OF CASCADED SOA-BASED WAVELENGTH CONVERTER

The configuration of the cascaded SOA-based wavelength converter is shown in Fig. 2(a). The wavelength converter consists of three polarization controllers (PCs), a polarizer, an optical bandpass filter (BPF), and a single SOA, which is one of the four SOAs, as shown in Fig. 1(c). Using the PCs at each signal output, the state of polarization (SOP) of the input data signal is adjusted to obtain the highest gain of the SOA, while the SOP of the probe beam is adjusted to be approximately 45° with respect to the axis of the SOA. When the data and probe beams are launched into the SOA, the probe beam experiences XGM and XPM, which causes polarization rotation of the probe beam due to a phase difference between the TE and TM modes of the SOA. Using the PC at the output of the SOA, the SOP of the probe beam is rotated so that minimum power is obtained at the polarizer output when the data signal is zero. In this way, NIV wavelength conversion based on nonlinear polarization rotation is performed [21]–[25]. Conversely, in this experiment, IV wavelength conversion is based on XGM by rotating the SOP of the probe beam so that a maximum power is obtained at the output of the polarizer. The BPF is employed to remove the residual data signal and suppress the amplified spontaneous emission (ASE) noise of the SOA. Fig. 2(b) shows the examples of the converted signal traces at 5.0 Gb/s for the case of the NIV and IV operations. The wavelengths of the input data and

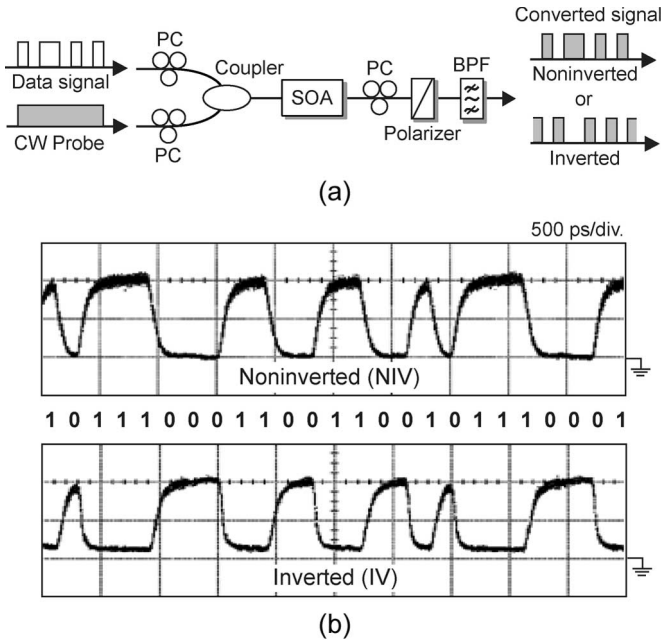


Fig. 2. (a) Configuration of cascaded SOA-based wavelength converter. (b) Converted pulse traces of 5.0-Gb/s NIV and IV wavelength conversions.

CW probe signals are 1540 and 1530 nm, respectively. It is clearly seen that these data codes of the traces are mutually inverted. In this scheme, operating speed is limited by the carrier recovery time of an employed SOA, since the rise and fall times of converted signals are determined by the recovery time. It should be noted that 40-Gb/s wavelength conversion using this scheme will be possible if we employ an SOA with faster recovery time [25].

In order to investigate input/output response of the cascaded converter, we measure the static characteristics of relative output power as a function of the input CW signal power for the two kinds of output wavelength. In this measurement, we examine the wavelength converter using SOA3 and the input data signal at the wavelength of 1540 nm. Fig. 3(a) shows the static characteristics of up-conversion to 1530 nm, which assumes a conventional wavelength conversion within high gain band. To obtain a good conversion performance, the injected probe power into SOA3 is set to 2.0 dBm. In the case of the NIV operation, the wavelength conversion is based on XPM, since the SOP of probe beam at the output of SOA is adjusted so that maximum XPM effect is obtained at “mark” signal level. On the other hand, the IV conversion requires a larger signal power to obtain deeper gain saturation at “mark” signal level. In general, XPM-based wavelength conversion can obtain a good conversion performance with a lower carrier density modulation compared to an XGM-based one [13]. Therefore a higher extinction ratio of this scheme is obtained with the NIV operation, as shown in Figs. 2(b) and 3(a).

The static characteristics of up-conversion of 1540 to 1460 nm are depicted in Fig. 3(b). This up-conversion with such a wide wavelength hopping range is applied to each wavelength converter for the cascaded broadband up-conversion, as shown in Fig. 1(b). The injected probe power of 8.0 dBm is adjusted to obtain the highest conversion performance. In Fig. 3(b), the relative output powers for both the NIV and IV operations are

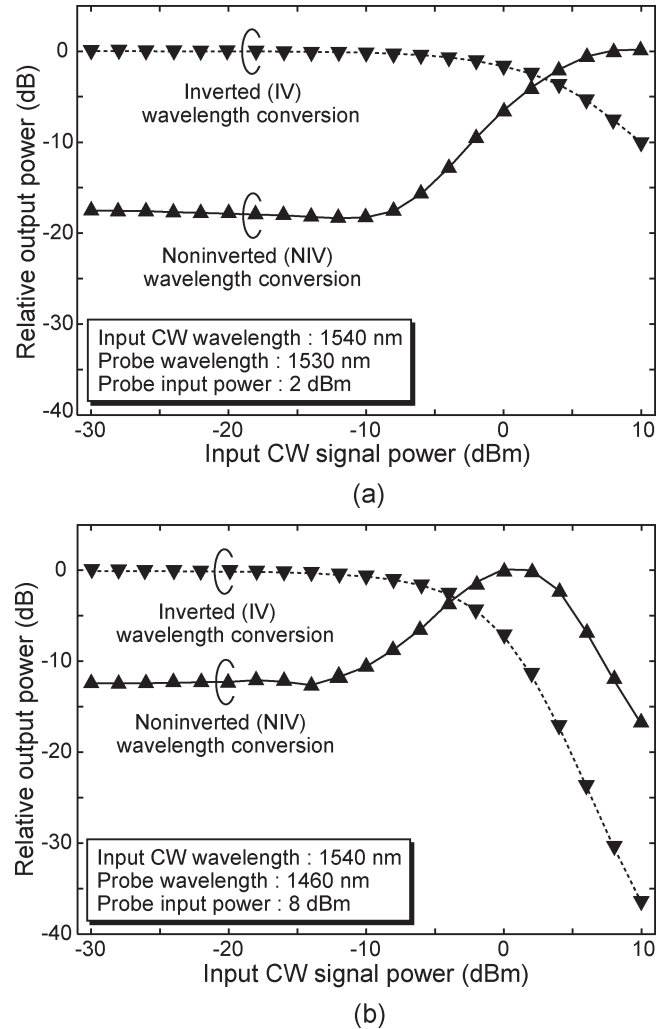


Fig. 3. Static characteristics of output power as a function of input CW signal power. (a) Conventional up-conversion of 1540 to 1530 nm. (b) Up-conversion of 1540 to 1460 nm for cascaded operation.

drastically decreased when the input CW signal power is over 2.0 dBm. This is due to the gain compression between the input CW and probe signal powers. The gain compression reduces the output power of the converted signal, which is originated from the probe signal, as the input CW probe power increases. This behavior is more remarkable when the probe wavelength is located at a shorter side of the gain peak wavelength. It gives a strong influence to conversion performance of the wavelength converter. As shown in Fig. 3(b), the gain compression opposes the effect of the NIV conversion but enhances the effect of the IV conversion. Therefore, the output extinction ratio of the IV operation is well improved, whereas the extinction ratio of the NIV-converted signal is degraded at a high-input power level.

Fig. 4 shows the output spectra of the static up-conversions at the output of SOA3. The injected probe powers for each up-conversion are the same as the static operations of Fig. 3. Here, we compare the effect of gain compression of the converted 1530- and 1460-nm signals. When the input CW signal power is changed from -5.0 dBm [Fig. 4(a)] into 10 dBm [Fig. 4(b)], the optical signal-to-noise ratio (OSNR) degradation of the 1460-nm signal is much larger than that of the 1530-nm signal. It should be noted that the spectral component at 1520 nm is an

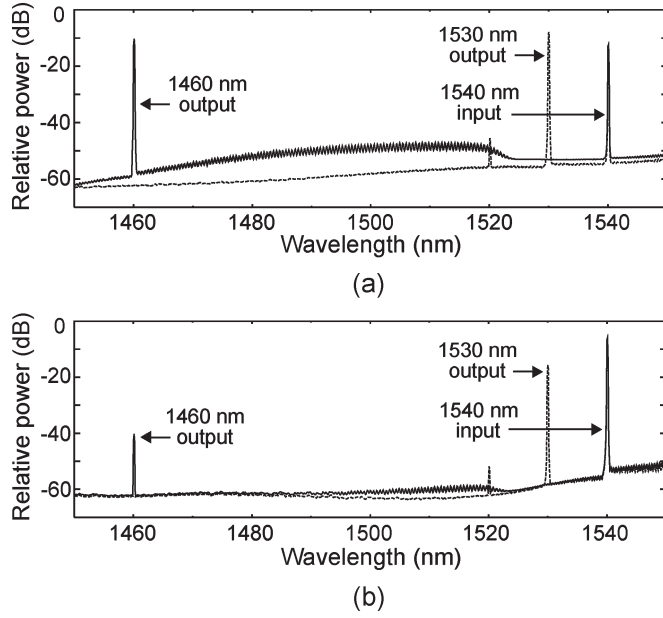


Fig. 4. Output spectra at the output of SOA3 for the cases of (a) -5.0 dBm and (b) 10 dBm of input CW signal powers. The 1460 - and 1530 -nm signals are plotted by solid and dashed lines, respectively.

idler component induced by four-wave mixing in SOA3. These results coincide well with the input/output characteristics, as shown in Fig. 3, and similar characteristics are also achieved for the other employed SOAs such as SOA2 and SOA4. This means that the gain compression is effective for improving the extinction ratio of the IV conversion for the cascaded scheme.

The measured extinction ratio after the IV conversion of 1540 to 1460 nm versus the input signal power is shown in Fig. 5. The operating bit rate is 5.0 Gb/s. The extinction ratio and the eye pattern, as shown in the insets, are measured by a digital sampling oscilloscope with 30 -GHz bandwidth. A maximum extinction ratio is obtained when the input data power is around 5 dBm. As the input data power becomes larger than 5 dBm, the input data power at “space” level is also increased, then the output power of the IV-converted signal at the “mark” level decreases and fluctuates. As a result, the extinction ratio of the converted signal degrades. In contrast, as the input power becomes smaller than 5 dBm, the output power of the IV-converted signal at the “space” level increases, and the extinction ratio degrades. These behaviors can be well understood by the input/output characteristics of the IV conversion in Fig. 3(b) and the eye patterns in the insets of Fig. 5.

Fig. 6 shows the wavelength dependence of the maximum extinction ratio and its input data signal power. Each of the injected CW probe powers for various input data powers is also optimized in the range of 2.0 – 8.0 dBm. As the wavelength of the converted signal shifts to a shorter side, the optimum input data power gradually becomes smaller, and the extinction ratio is sharply degraded. This indicates that the input data power has to be optimized for the wavelength of the converted signal, although high extinction ratio can be obtained in a wide wavelength range. It should be noted that we cannot measure the extinction ratio of the converted signal at the wavelength range of 1435 – 1445 nm, since the range is outside of the operating

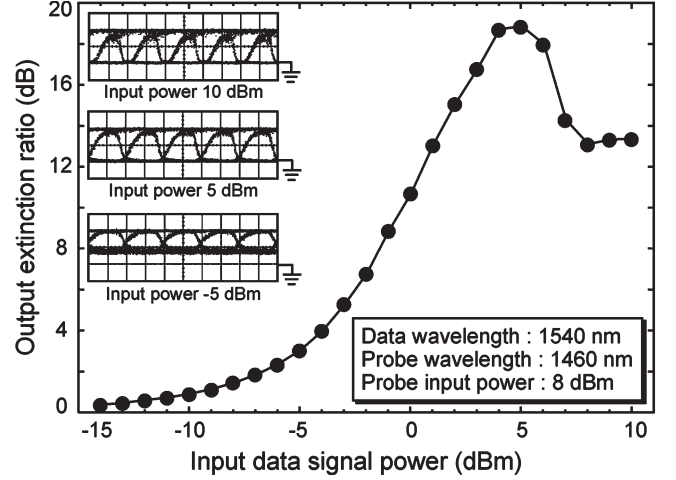


Fig. 5. Measured extinction ratio after the conversion of 1540 to 1460 nm versus input data signal power. Insets show the eye patterns of the converted signals for various input signal powers.

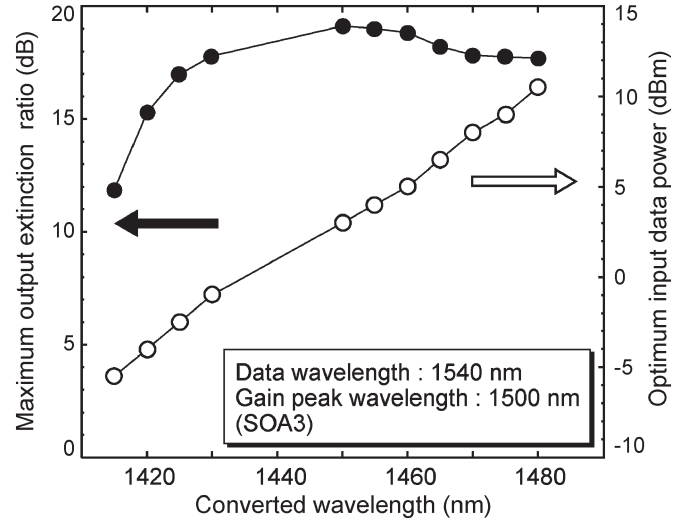


Fig. 6. Maximum extinction ratio and its input data signal power versus converted wavelength of output signal for the cascaded up-conversion.

wavelength of the employed E- and S-band wavelength-tunable light sources.

IV. EXPERIMENT AND RESULTS

A. Experimental Setup

The experimental setup for ultrawideband wavelength conversion using the single or multistage cascaded wavelength converter is shown in Fig. 7. These wavelength converters utilize four SOAs, as shown in Fig. 1(c). As transmitters, the 5.0 -Gb/s NRZ signals at wavelengths of 1625 , 1540 , and 1320 nm are employed. These wavelengths are determined so that the cascaded conversion with larger hopping range and higher quality is performed by using our experiments. The NRZ signals are generated by a 1.3 - or 1.55 - μm LiNbO₃ modulator and coded into a pseudorandom bit sequence with a $2^{31} - 1$ -bit pattern length. The CW probe lights are also generated by wavelength-tunable laser diodes (TLDs) with each various wavelength-tunable range. As preamplifiers for the injected

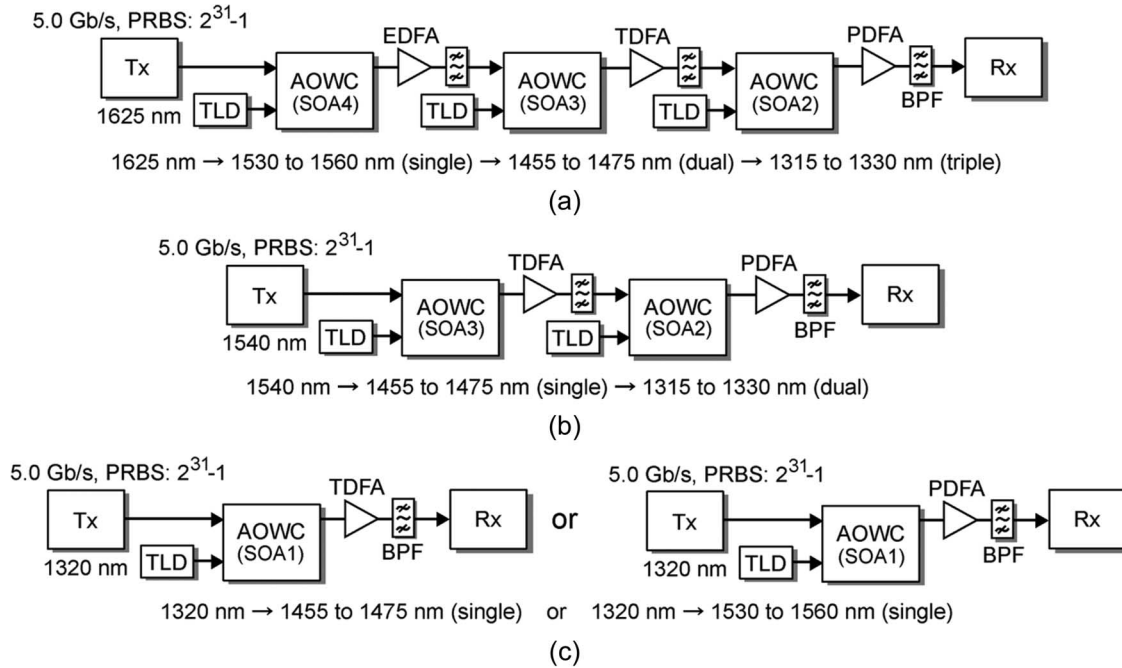


Fig. 7. Experimental setups for the ultrawideband wavelength conversion. (a) Up-conversion of 1625-nm signal to short wavelengths using single-, dual-, or triple-stage cascaded wavelength converter. (b) Up-conversion of 1540-nm signal using single- or dual-stage wavelength converter. (c) Down-conversions of 1320 nm to long wavelengths using two types of single-stage converter.

data signals into each converter, an erbium-doped fiber amplifier (EDFA), a thulium-doped fiber amplifier (TDFA), and a praseodymium-doped fiber amplifier are employed in each amplified signal wavelength region. The injection currents of the employed SOAs (SOA1, SOA2, SOA3, and SOA4) are adjusted to obtain proper gain, and these values are 200, 120, 200, and 500 mA, respectively. The SOA1, SOA3, and SOA4 are polarization independent and commercially available fiber-pigtailed components, whereas the SOA2 employs a conventional Fabry-Pérot laser diode with antireflection coating and strong polarization sensitivity. In Fig. 7(a) and (b), up-conversions of 1625- and 1540-nm signals are performed by the single-, dual-, or triple-stage cascaded converter corresponding to each wavelength hopping range. On the other hand, as shown in Fig. 7(c), down-conversion is performed by the single-stage SOA-based wavelength converter based on nonlinear polarization rotation [24]. In these experimental setups, the injected powers of the data and probe signals into each SOA are set to 5 to 8 dBm. These values are adjusted so that the highest qualities of the converted signals are obtained in each wavelength converter. To estimate conversion performances of our proposed scheme, we measure bit error rates (BERs) and Q -factors of the converted signals using a BER tester at each receiver.

B. Regeneration Effect of Cascaded Wavelength Converter

In the case of cascaded broadband up-conversion, it is thought that the quality of converted signal is degraded as the number of cascaded converters is increased. However, it is possible to suppress the degradation of signal quality if the input/output characteristics, as shown in Fig. 3(b), are effectively utilized. Here, in order to compare the signal qualities of the

back-to-back and converted signals, we investigate the power penalty ($\text{BER} = 10^{-9}$) of the output signal with changing the OSNR of input data signal.

The experimental setup is shown in Fig. 8(a). The injected data signal power into the converter is set to 2.0 dBm for all the setups. To control the OSNR of the input data signal, we employ an ASE noise source, which consists of a variable optical attenuator and dual-stage EDFA, which is at the input of the wavelength converter using the single- or dual-stage cascaded scheme, as shown in Fig. 7(b). The wavelength and bit rate of the input data signal are 1540 nm and 5.0 Gb/s, respectively. Fig. 8(b) shows the input OSNR tolerances of the converted signals. For all the results, the power penalties become large as the OSNR of the input signal decreases. On the other hand, in the case of single-stage conversion to 1460 nm, the power penalty of the IV-converted signal can be kept lower compared to the back-to-back and NIV-converted signals, even when the input OSNR is degraded. Moreover, repeated IV conversion (conversion of 1540 to 1320 nm) further improves the OSNR tolerance. The reason is as follows. In the single-stage IV conversion to 1460 nm, the higher extinction ratio is obtained due to the gain compression between input data and probe signals, as shown in Fig. 3(b). This behavior converts large input power fluctuation at “mark” level into very small output fluctuation at “space” level. Moreover, the repeated IV conversion simultaneously suppresses the output fluctuations at “mark” and “space” levels. This results from not only the improved extinction ratio but also from the flat input/output response in a small signal level over a wide input range at the IV conversion, as shown in Fig. 3(b). Thus, the output amplitudes of the converted signal at both “mark” and “space” levels are regenerated by the dual-stage cascaded IV conversion. These effects can also be seen from the output eye patterns

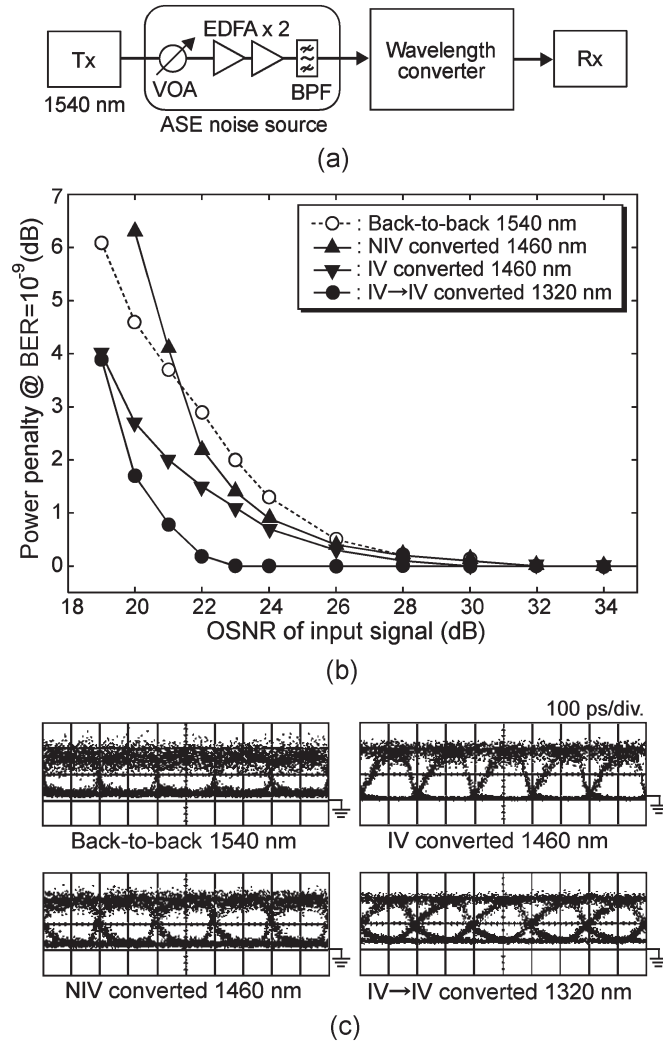


Fig. 8. (a) Experimental setup. (b) Measured input OSNR tolerances of back-to-back and converted signals using single- and dual-stage wavelength converters. (c) Eye patterns of the output signals when input OSNR is 22 dB/0.1 nm.

when the input OSNR is 22 dB with 0.1-nm resolution, as shown in Fig. 8(c). On the other hand, we obtain a lower extinction ratio of the 1320-nm signal than that of the 1460-nm IV converted signal. This is due to the poor gain characteristics of SOA2 for obtaining a high gain compression effect in the large wavelength hopping range of 1460–1320-nm conversion. If we employ a proper SOA with higher gain and a shorter gain peak wavelength, we will obtain a higher extinction ratio of the 1320-nm signal.

C. Ultrawideband Wavelength Conversion Over 300 nm

To examine up-conversion performance using cascaded wavelength converter, we measure the signal spectrum of CW probe light at the output of the SOA in each converter. The output spectra of the single-stage converter using SOA4 are shown in Fig. 9(a). The output OSNRs of the probe lights at 1320 and 1460 nm are drastically degraded due to the high absorption loss of SOA4. It should be noted that the two spectra around 1445 nm are spurious components that are not induced by the probe light but observed by the employed

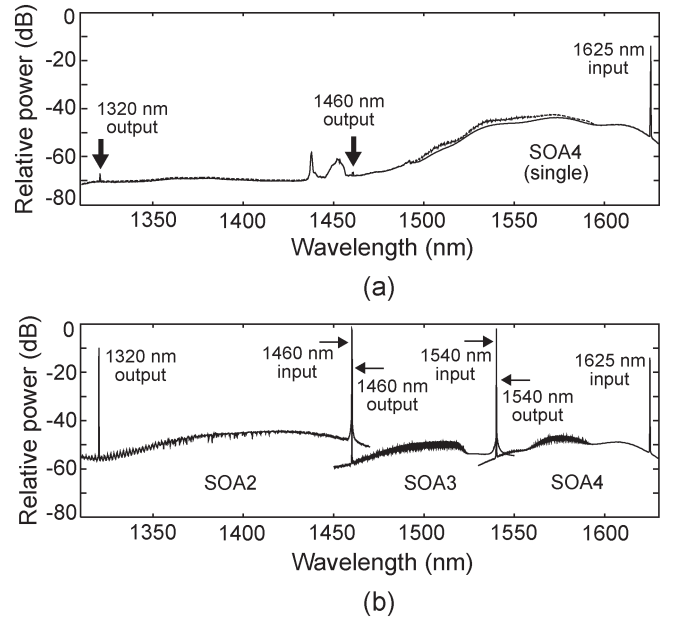


Fig. 9. Output spectra at the output of the polarizer in each wavelength converter. (a) Single-stage up-conversions of 1625 to 1460 and 1320 nm. (b) Triple-stage up-conversion of 1625 to 1320 nm.

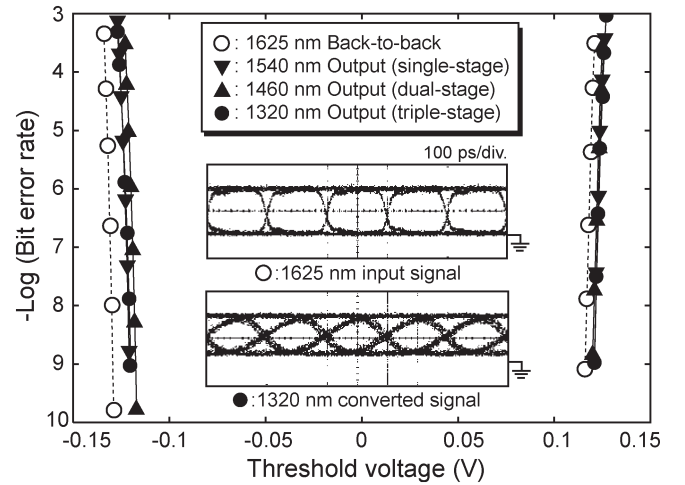


Fig. 10. Measured BERs versus threshold voltages. Insets show the eye patterns of back-to-back signal at 1625 nm and converted signal at 1320 nm.

spectrum analyzer and the 1625-nm light source. In contrast, the triple-stage cascaded up-conversion, as shown in Fig. 9(b), preserves high-output OSNRs at much shorter wavelength side. In addition, we also demonstrate tunable operations of the converted wavelength from 1540 to 1625, 1460 to 1540, and 1320 to 1460 nm, respectively. Hence, the triple-stage cascaded converter enables us to obtain high-output OSNR over 300-nm wavelength range.

Fig. 10 shows the measured BERs versus the threshold voltages of the O/E converted signals for the case of the cascaded up-conversion of 1625 to 1540 nm (single stage), 1460 nm (dual stage), and 1320 nm (triple stage). All of the cascaded conversions employ the IV operation. The comparable threshold-voltage characteristics are achieved with the back-to-back windows even if the up-conversion is repeated. The insets show the eye patterns of the back-to-back signal and the

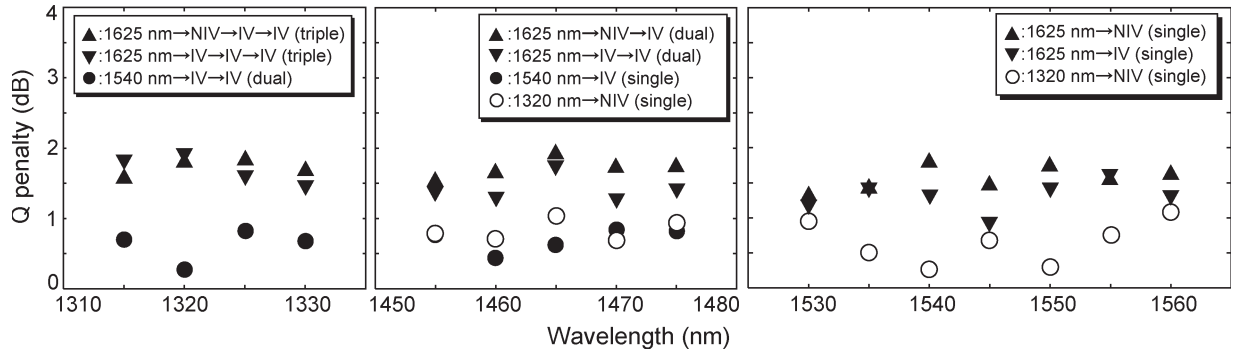


Fig. 11. Measured Q -penalties of the converted signals using single-, dual-, and triple-stage wavelength conversions for various output wavelengths. These conversions are performed at NIV and IV operations.

converted signal using the triple-stage cascaded converter. In spite of multiple cascaded wavelength conversion, the eye pattern shows a clear eye and indicates good conversion performance, owing to amplitude regeneration effect as shown in Fig. 8. On the other hand, we observe the slow rise and fall times of the converted signal. This is due to the slow carrier recovery times of the employed SOAs in each cascaded converter.

Next, in order to estimate the signal quality of the converted signal using our proposed scheme, we measure the Q -factors of the converted signals in the wavelength range of 1315–1625 nm. The Q -factors are interpolated from the inverse BER function [26]. The measured Q -penalties for various output wavelengths are shown in Fig. 11. In this experiment, we perform the IV conversion for the cascaded operation, because better conversion performance is achieved when the IV conversion is repeated, as shown in Fig. 8. On the other hand, the first stage converter in the triple-stage cascaded scheme is performed at both the NIV and IV operations to compare the signal qualities of the converted signals with both logic outputs for the dual- and triple-stage cascaded conversion. The Q -values of the back-to-back signals at 1320, 1540, and 1625 nm are 36.97, 37.50, and 37.44 dB, respectively. For all the output wavelengths, the low Q -penalties less than 2.0 dB are achieved. Moreover, no remarkable Q -degradation of the converted signal is observed even if the up-conversion is repeated. This indicates that the cascaded wavelength conversion preserves high signal quality, and the amplitude regeneration effect is also effectively utilized for the ultrawideband wavelength conversion. We also demonstrate the cascaded operation by the repeated NIV conversion. However, the comparable signal qualities to the cascaded IV conversion are not obtained due to high signal degradation.

V. DISCUSSION

In this paper, we demonstrated ultrawideband wavelength conversion using a multistage cascaded wavelength converter. The achieved wavelength hopping range of 310 nm is the largest value reported to date, to the best of our knowledge. Since the proposed scheme is based on XGM and XPM in SOA with a simple technique, which repeats up-conversion using multiple SOAs each with different gain band, it is applicable to any type of SOA if employed SOAs have proper gain characteristics in the wide wavelength range. On the other hand, available

operation speed is limited by the carrier recovery times of employed SOAs. Although the bit rate of our presented experiments was carried out at 5.0 Gb/s due to the slow recovery time of the employed SOAs, especially SOA2, the ultrawideband wavelength conversion over 10 Gb/s will be possible using proper SOAs with faster recovery time.

By repeating the IV conversion, we also demonstrate the regeneration effect of the cascaded wavelength converter. This technique plays an important role in not only the performing all-optical reamplifying and reshaping (2R) regenerator with broadband up-conversion but also improving the cascadability of the proposed approach. In this experiment, the amplitude of the converted signal is well regenerated. Therefore, further improvement of the operating and hopping wavelength ranges, which obtain high-quality signal, will be expected, using more cascaded converters with each proper gain band of the SOAs.

VI. CONCLUSION

We have demonstrated a novel scheme to achieve an all-optical and ultrawideband wavelength conversion over a 300-nm wavelength hopping range. The original point in our approach is to utilize multistage cascaded SOA-based wavelength converters, each with a different gain band. The essential feature of the employed wavelength converter is to perform both NIV and IV wavelength conversions. In particular, the IV wavelength conversion plays an important role in preserving a high signal quality for cascaded operation. We have also shown an amplitude regeneration effect and better cascadability of our proposed scheme by repeating IV conversion. Moreover, ultrawideband wavelength conversions with low Q -penalties less than 2.0 dB are achieved in the wavelength range of 1315–1625 nm. In our proposed technique, further improvements of operating bit rate and wavelength range will be possible using more cascaded wavelength converters by SOAs with faster recovery time. These results indicate that our proposed technique will be useful in improving the flexibility of future ultrawideband photonic networks.

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- Motoharu Matsuura**, photograph and biography not available at the time of publication.
- Naoto Kishi** (S'85–M'87), photograph and biography not available at the time of publication.
- Tetsuya Miki** (S'70–M'70–SM'95–F'00), photograph and biography not available at the time of publication.